

Math142 - Practice Problems for Exam 1

1. Evaluate the following integrals.

(a) $\int_0^{\pi/2} \frac{\cos \theta}{1 + \sin \theta} d\theta$

(b) $\int \frac{\ln x}{x^2} dx$

(c) $\int y e^{-2y} dy$

(d) $\int \cos \sqrt{t} dt$

2. Evaluate the following integrals.

(a) $\int \sin^3 \theta \cos^2 \theta d\theta$

(b) $\int \tan^5 \theta \sec^3 \theta d\theta$

(c) $\int \cos^2 \theta d\theta$

(d) $\int_1^2 \frac{\sqrt{x^2 - 1}}{x} dx$

(e) $\int \frac{x - 9}{x^2 + 3x - 10} dx$

3. Evaluate each of the following improper integrals or show that it is divergent.

(a) $\int_2^\infty \frac{dx}{x \ln x}$

(b) $\int_2^6 \frac{dy}{\sqrt{6 - y}}$

4. Find the area of the region bounded by the given curves.

(a) $y = \sqrt{x}$, $y = x^2$.

(b) $y = 2 - x^2$, $y = |x|$.

5. Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified lines.

(a) $y = 2x$, $y = x^2$; about the x -axis.

(b) $y = \ln x$, $y = 1$, $y = 2$, $x = 0$; about the y -axis.